Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

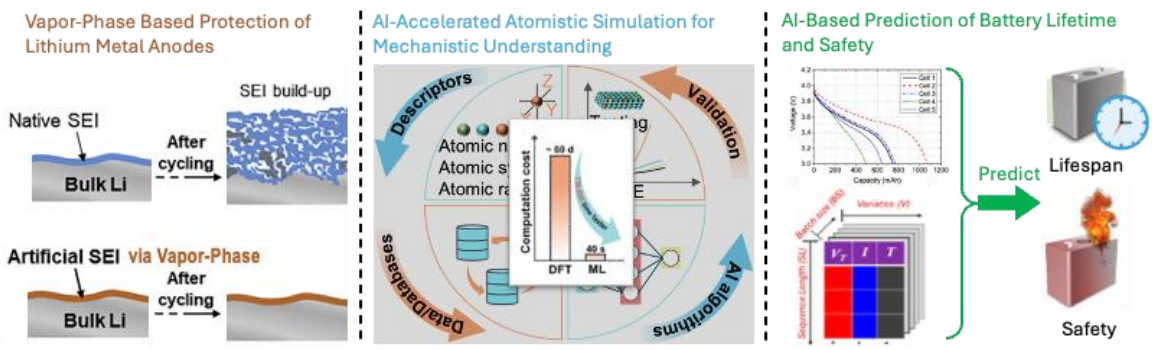
Title of Proposed Technology	**AI-guided CVD polymer interphases for uniform lithium plating and interfacial stability in high-energy-density lithium metal batteries**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		*Chemicals*	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	*Secondary Batteries / Battery Materials*	*Polymeric Materials*	*Polymer Processing Technologies*
Concept of Proposed Technology	**Core Concept** This technology proposes the integration of vapor-phase–derived polymer interphase engineering with AI-assisted analysis establishes a scalable and broadly applicable pathway to regulate lithium deposition and enhance the safety and reliability of lithium metal batteries. 			

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	Figure 1. Vapor-phased-enabled lithium metal anode stabilization integrated with AI-accelerated mechanistic understanding and predictive battery diagnostics. **Key Functions and Potential Applications** 1) Vapor-Phase–Derived Mixed Electron/Ion-Conductive Polymer Interphases ● This project proposes an interfacial engineering strategy based on mixed electron/ion-conductive polymer interphases formed via vapor-phase polymerization. The solvent-free, vapor-phase process enables ultrathin and conformal polymer layers to be directly formed on reactive lithium metal surfaces, allowing precise control over interfacial chemistry, thickness, and transport properties. By simultaneously regulating electronic leakage and lithium-ion transport at the anode interface, the polymer interphases are designed to homogenize lithium nucleation and growth, suppress dendritic deposition, and improve long-term interfacial stability under practical operating conditions. 2) AI-Assisted Mechanistic Understanding and Safety Diagnostics ● Artificial intelligence is integrated as an enabling tool to accelerate mechanistic analysis and safety evaluation. AI-assisted surrogate modeling expedites DFT and AIMD-level exploration of polymer–lithium–electrolyte interactions, while data-driven analysis of electrochemical signals enables early identification of instability signatures associated with internal short-circuit risk. Importantly, AI augments physics-based modeling and experimental validation, rather than replacing them.
Rationale for Joint R&D Project	**Critical Technical Barriers** Most existing interfacial engineering strategies for lithium metal anodes rely on conventional solution-based coating methods, such as spray, dip, and spin coating. While straightforward to implement, these approaches are fundamentally incompatible with the high reactivity and mechanical softness of lithium metal. Unavoidable surface tension and solvent–substrate interactions lead to poor wetting, non-uniform coverage, local dewetting, and defect formation that expose bare lithium (Figure 2). In addition, solvent contact can trigger parasitic reactions and surface roughening, further amplifying interfacial heterogeneity. As a result, solution-based methods struggle to produce ultrathin, conformal, and chemically uniform interphases on lithium metal anodes. As a consequence, existing coating strategies provide limited control over interfacial chemistry, thickness uniformity, and mechanical integrity during repeated lithium plating and stripping. This lack of precise interfacial control has hindered systematic understanding of how purpose-built artificial interphases regulate lithium nucleation, deposition morphology, and long-term interfacial stability. Overcoming these limitations requires solvent-free interfacial engineering approaches capable of forming uniform and stable coatings directly on lithium metal anodes.

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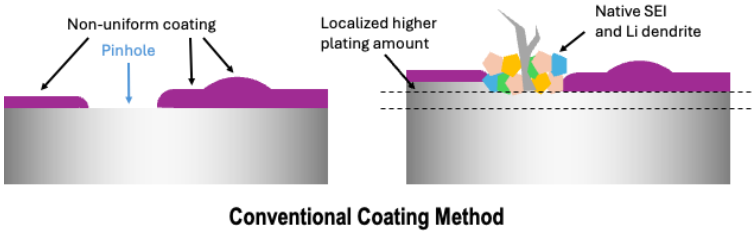

Figure 2. Limitations of conventional coating methods for Li metal anode.

Why Global Collaboration?

1) Vapor-Phase Polymerization as a Scalable Interfacial Engineering Platform

To address these challenges, vapor-phase polymerization offers a solvent-free and manufacturing-relevant alternative for interfacial engineering. By eliminating liquid-phase transport and wetting constraints, vapor-phase processes enable the formation of uniform, conformal polymer interphases with precise control over chemistry, thickness, and interfacial uniformity, even on highly reactive lithium metal surfaces. As such, vapor-phase polymerization represents a promising and scalable platform for regulating lithium deposition behavior and improving interfacial stability in practical lithium metal battery systems.

2) Need for Integrated Interfacial and Device-Level Development

Translation of vapor-phase–engineered interphases to realistic battery applications requires systematic validation under device-relevant conditions, including larger electrode areas, higher areal capacities, and extended cycling. Achieving this translation necessitates integrated development that couples interfacial engineering with device-level fabrication, electrochemical testing, and safety evaluation. Such integration is essential to assess whether interfacial benefits observed at the materials and coin-cell levels can be reliably maintained in larger-format cells and under practical operating conditions.

3) Complementary Academic–Industrial Capabilities

Within the proposed joint R&D framework, the proposing team contributes core capabilities in vapor-phase polymerization–based interfacial engineering, including polymer chemistry design, interphase formation, and mechanism-oriented evaluation of lithium deposition behavior. To enable practical translation, complementary capabilities are required from an industrial partner, particularly in device-relevant lithium metal cell fabrication, intermediate- to large-format cell assembly, and performance and safety evaluation under application-relevant conditions. In addition, access to structured electrochemical datasets and experience in data-driven diagnostics are essential for validating interfacial designs and identifying early-stage instability signatures at the device level.

Consequently, through iterative feedback linking interfacial design, mechanistic understanding, device-level validation, and electrochemical diagnostics, this joint R&D collaboration establishes an efficient and scalable pathway for advancing vapor-phase–engineered lithium metal anodes from laboratory concepts toward safety-aware, application-ready battery technologies.

Key Words	1. Li metal anode 2. Vapor-phase polymerization 3. Artificial SEI

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	4. Machine learning 5. Battery safety

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	**R&D Objectives** To develop a scalable and safety-oriented lithium metal anode by regulating lithium deposition through vapor-phase–engineered polymer interphases, and to validate its performance under device-relevant conditions through integrated interfacial design, modeling, and cell-level evaluation. **Detailed Objectives** 1) Develop and optimize vapor-phase–derived mixed electron/ion-conductive polymer interphases for uniform lithium deposition and stable interfacial behavior. This objective focuses on achieving precise and reproducible interfacial control on highly reactive lithium metal surfaces. By integrating polymer chemistry design with vapor-phase interfacial formation, the project aims to construct ultrathin and conformal polymer interphases that directly regulate lithium nucleation and growth behavior, thereby suppressing morphological instability during repeated plating and stripping. 2) Establish a mechanistic understanding of lithium nucleation and growth at polymer-modified lithium metal interfaces using AI-accelerated atomistic simulations. First-principles–based simulations will be used to analyze lithium adsorption, diffusion, and nucleation behavior at polymer–lithium interfaces. Key interfacial descriptors derived from these simulations will be used to train supervised learning models that capture the relationship between polymer chemistry, interfacial structure, and lithium nucleation behavior. The resulting surrogate models enable rapid evaluation of nucleation and growth trends across an expanded interfacial design space and provide guidance for interphase optimization. 3) Device-Level Validation of Vapor-Phase Modified Lithium Metal Anodes in Ah-Class Pouch Cells This objective focuses on translating optimized polymer interphase designs from electrode-level studies to device-relevant battery configurations. Polymer-modified lithium metal anodes will be integrated into intermediate- to large-format cells to evaluate electrochemical performance, interfacial stability, and scalability under realistic operating conditions. This effort establishes the device-level relevance of the proposed interfacial strategy and assesses its applicability to high-energy lithium metal batteries. 4) AI-Based Electrochemical Diagnostics for Early Detection of Internal Short-Circuit Risk This objective aims to develop AI-assisted diagnostic methods for safety-oriented assessment using electrochemical operation data. Physics-informed features extracted from voltage, capacity, relaxation, and impedance signals will be used to train anomaly-detection models that establish normal cell behavior and identify early deviations associated with interfacial instability and internal short-circuit risk. The resulting framework enables interpretable, non-destructive safety assessment that complements device-level validation.			

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R&D Target Specifications	● A controlled compositional library comprising four to five representative conductive polymer systems will be established to systematically evaluate the influence of polymer chemistry on lithium interfacial behavior. ● Vapor-phase interfacial formation parameters will be systematically optimized to achieve uniform deposition, high interfacial conformity, and electrochemical stability on highly reactive lithium metal surfaces. ● AI-assisted atomistic modeling will be employed to interpret lithium deposition and interfacial stabilization mechanisms, establishing causal relationships between polymer interphases and dendrite suppression. ● The effectiveness of the polymer interphases in regulating lithium plating and stripping will be benchmarked through stable symmetric-cell cycling at high current density, targeting at least a 150% improvement over unmodified electrodes. ● The applicability of the interfacial strategy to high-energy-density configurations will be demonstrated using ultrathin lithium metal anodes with thicknesses of 20 μm or below while maintaining high-capacity retention over extended cycling. ● Device-level scalability will be validated through demonstration in intermediate-format pouch cells incorporating optimized polymer interphases. ● Meaningful performance translation from small-scale cells to pouch-cell configurations will be verified by achieving at least a 150% improvement in cycle life at practical charge–discharge rates compared to baseline cells. ● AI-assisted electrochemical diagnostics will be developed to enable early, non-destructive detection of internal short-circuit–related anomalies based on electrochemical operation signals prior to catastrophic failure.
R&D Plan	**Year 1 (6 months): Project Initiation and Baseline Establishment** 1) Establishing and validating vapor-phase interfacial formation workflows to achieve uniform and conformal polymer coatings on highly reactive lithium metal surfaces. 2) Fabricating and evaluating coin-cell configurations incorporating polymer-modified lithium metal anodes to establish baseline lithium deposition behavior and optimize interfacial formation parameters. 3) Establishing baseline lithium metal cell configurations and standardized electrochemical testing protocols under device-relevant conditions. 4) Initiating AI frameworks for electrochemical diagnostics by defining target electrochemical signatures associated with interfacial instability and internal short-circuit–related anomalies. **Year 2 (12 months): Interfacial Optimization and Mechanistic Elaboration** 1) Performing systematic coin-cell-level electrochemical testing to evaluate lithium plating uniformity, interfacial resistance evolution, and cycling stability as a function of vapor-phase interfacial formation parameters. 2) Validating the stability of ultrathin lithium metal anodes (≤20 μm) incorporating optimized polymer interphases at the coin-cell level.

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3) Establishing mechanistic understanding of lithium nucleation, growth, and interfacial stabilization through integrated analysis of experimental observations and AI-accelerated atomistic modeling results.
4) Refining polymer chemistry and vapor-phase interfacial formation parameters based on experimental and modeling insights to guide interphase optimization.
5) Generating structured electrochemical datasets capturing both stable operation and early-stage degradation behavior to support diagnostic model development.
6) Training and refining AI-based electrochemical diagnostic models using accumulated cell-level data, with emphasis on interpretability and robustness.

Year 3 (12 months): Device-Level Integration and Safety-Oriented Evaluation

1) Integrating optimized polymer-modified lithium metal anodes into intermediate-format cells and conducting extended electrochemical evaluation.
2) Assessing interfacial compatibility between optimized polymer interphases and representative electrolyte systems, focusing on stability and transport behavior.
3) Consolidating mechanistic understanding of lithium deposition and interfacial evolution through cross-analysis of cell-level data and atomistic modeling.
4) Conducting pouch-cell-level cycling and diagnostic testing to evaluate performance, scalability, and interfacial durability under realistic operating conditions.
5) Applying AI-assisted analysis to device-level electrochemical data to identify early-stage internal short-circuit–related anomalies and degradation trends.

Year 4 (6 months): Predictive Safety Validation and Technology Demonstration

1) Linking electrochemical instability signatures observed in device-level testing to interfacial mechanisms identified in earlier stages.
2) Finalizing interfacial design principles and vapor-phase interfacial formation parameter guidelines applicable to practical lithium metal battery systems.
3) Validating AI-based electrochemical diagnostic models for predictive identification of internal short-circuit–related anomalies using accumulated device-level datasets.
4) Demonstrating polymer-modified lithium metal pouch cells integrated with AI-assisted predictive safety assessment.
5) Summarizing project outcomes in terms of electrochemical performance, safety behavior, and readiness for further scale-up and commercialization.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	To lead interfacial engineering and mechanism-oriented research for vapor-phase–engineered lithium metal anodes, providing materials design, coin-cell validation, and physics-informed feedback to support device-level development and AI model refinement. 1) Design and evaluate mixed electron/ion-conductive polymer systems for lithium metal interfaces, with emphasis on interfacial stability and lithium deposition regulation. 2) Develop and optimize solvent-free vapor-phase interfacial formation processes to achieve uniform and conformal polymer interphases on reactive lithium metal surfaces. 3) Perform coin-cell-level electrochemical testing to assess lithium plating/stripping behavior and validate interfacial performance prior to scale-up. 4) Establish structure–property–behavior relationships linking polymer chemistry, interfacial characteristics, and lithium deposition stability. 5) Provide mechanism-based insights and physics-informed feedback to support refinement of AI-assisted electrochemical diagnostic models and contribute to intellectual property development related to interfacial design strategies.
Expected R&R of Korean SME Partner	To lead device-level implementation, cell fabrication, electrochemical testing, and AI-assisted diagnostics, enabling translation of interfacial innovations into practical lithium metal battery performance and safety evaluation. 1) Fabricate lithium metal cells across relevant formats, including coin and intermediate-format pouch cells, to support device-level validation under application-relevant conditions. 2) Conduct systematic electrochemical testing to evaluate cycling stability, rate capability, durability, and scalability of polymer-modified lithium metal anodes. 3) Develop and implement AI-assisted electrochemical diagnostic models for analyzing operational data and identifying early-stage internal short-circuit–related anomalies and safety-relevant instabilities. 4) Establish and manage electrochemical data acquisition, preprocessing, and integration pipelines to support robust model training and cross-scale analysis. 5) Integrate experimental results with AI-based safety assessment and contribute to scale-up evaluation, intellectual property generation, and preparation for technology transfer and commercialization.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technology and System-Level Outcomes** 1) Establish a scalable vapor-phase interfacial engineering approach to improve lithium metal anode stability under practical operating conditions. 2) Demonstrate device-level feasibility of the proposed interfacial strategy in realistic battery configurations. 3) Integrate interfacial design with electrochemical analysis to enable systematic performance and stability evaluation. **Original Technical Outcomes** 1) Provide mechanistic insight into lithium deposition regulated by engineered interphases. 2) Establish interface-sensitive electrochemical evaluation and predictive diagnostic capability. 3) Form a closed-loop framework linking experiments, mechanistic understanding, and diagnostics. **Broader Impact and Adoption Potential** 1) Address key barriers to the practical deployment of lithium metal batteries. 2) Support industrial battery development through device-relevant evaluation and safety-oriented diagnostics. 3) Enable broader adoption of AI-assisted electrochemical analysis in battery testing workflows.

6. References

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